

Yanwei (Eric) Feng

(951) 813-1669 | yanweife@usc.edu | linkedin.com/in/yanwei-feng-284295288 | FinPilot Demo |

Los Angeles, CA

PROFESSIONAL SUMMARY

USC ISE / UCSD Applied Math graduate building production LLM systems for financial reasoning and decision-support AI -- domain-specific fine-tuning (LoRA), end-to-end data pipeline design, evaluation frameworks, and AWS deployment. Deployed a financial AI assistant on AWS EC2 with LoRA-fine-tuned domain reasoning; built complete MLOps pipelines from raw data ingestion through training, evaluation, and retraining. Designed safety guardrails and evaluation frameworks across financial, medical, and equity research AI systems where model errors have real consequences. Demo: 18.217.220.60

CORE COMPETENCIES

Financial Domain LLM Systems

LLM Fine-tuning (LoRA/PEFT)

Production Data Pipelines & MLOps

AI Evaluation Frameworks & Safety Guardrails

Multi-Agent Orchestration

AWS Deployment & AI Lifecycle Management

Python & SQL Systems

Prompt Engineering & Output Evaluation

WORK EXPERIENCE

Shelley Whitman Associates (SWA)

2024

Project Manager Intern -- Los Angeles, CA

- Coordinated cross-functional teams (design, engineering, marketing) to align deliverables with client requirements and resolve production bottlenecks before release
- Managed project timelines and resource allocation across parallel client engagements using Microsoft Project; maintained milestone checkpoints and delivery schedules
- Identified automation opportunities in scheduling and communication workflows; translated stakeholder requirements into actionable production scope

PROJECTS

Financial Decision Assistant (FinPilot) Live Demo & AWS EC2

Scoped 4 financial reasoning scenarios (market interpretation, risk analysis, strategy reasoning, scenario analysis) from user friction with general-purpose LLMs; cut 2 features with documented tradeoffs to ship v1 in 4 weeks. **Fine-tuned Qwen 0.6B with LoRA/PEFT for domain-specific financial reasoning**; deployed on AWS EC2 with multi-turn conversation logging and context management. Designed structured prompt templates for consistent financial reasoning outputs and evaluated model quality against domain-specific accuracy benchmarks. Live at 18.217.220.60.

Python · Qwen 0.6B · LoRA/PEFT · HuggingFace Transformers · AWS EC2 · Multi-turn Context Management · Financial Domain AI

Citation-Aware Equity Research Assistant Financial RAG + Evaluation

Built a full-stack financial research system with **FastAPI backend + Streamlit frontend**; integrated Polygon.io market data, NewsAPI, pandas-ta technical indicators, and **SEC filing RAG via ChromaDB + EDGAR** embeddings. Designed a **deterministic confidence score (0-100) computed from data completeness, signal coherence, and evidence depth** -- not LLM self-report -- as a safety guardrail that blocks insufficiently supported reports. Implemented thesis back-testing vs SPY with return, drawdown, and Sharpe metrics.

Python · FastAPI · Streamlit · ChromaDB · GPT-4o · Polygon.io · SEC EDGAR RAG · Deterministic Confidence Gate · pandas-ta

Financial LLM Training Pipeline End-to-End MLOps

Built an **end-to-end MLOps pipeline covering the full AI lifecycle**: raw data ingestion (finance-alpaca + synthetic reasoning data) -- data filtering and quality control -- JSONL transformation and train/val split -- LoRA fine-tuning with gradient accumulation and evaluation checkpoints -- model evaluation on validation samples. Designed for reproducibility and scalability; pipeline re-runs produce consistent training artifacts with logged metrics at every stage.

Python · HuggingFace Transformers & Datasets · LoRA/PEFT · PyTorch · JSONL Pipelines · MLOps · Experiment Tracking

AI Job Search Automation System Production Data Pipeline

Production data pipeline with multi-agent orchestration: ingests 500+ structured documents via parallel LLM workers, applies A-G weighted evaluation scoring (10 dimensions), and generates structured decision-ready reports with stateful execution -- dedup logic and error recovery at every stage, zero data integrity failures. Evaluation framework design: confidence-gated outputs block low-quality results before reaching the user; same architectural pattern as ML safety guardrails in production AI systems.

Python · Anthropic Claude API · Multi-Agent Orchestration · Tool-Calling Pipelines · Evaluation Framework · Stateful Execution

Medical QA System (Zhima Program) Safety Guardrails & Eval

Designed an **evaluation framework for high-stakes domain AI**: hallucination detection, output scoring rubrics, and confidence-gate logic to block factually incorrect medical responses before they reach users. Applied prompt constraints, context-window management, and dataset design to measurably improve factual accuracy -- same reliability-first approach required for regulated industry AI deployment.

Python · LLM APIs · Evaluation Framework Design · Hallucination Detection · Domain Prompt Engineering · Safety Guardrails

EDUCATION

M.S. Industrial & Systems Engineering -- University of Southern California (USC)

2024-2026

GPA: 3.82 / 4.0 | Coursework: Optimization & Convex Analysis, Machine Learning, Time Series Analysis, Numerical Methods

B.S. Applied Mathematics -- University of California, San Diego (UCSD)

2022-2023

GPA: 3.6 / 4.0

SKILLS

AI / LLM: LLM fine-tuning (LoRA/PEFT), HuggingFace Transformers, prompt engineering, evaluation frameworks & safety guardrails, LLM API integration (Anthropic Claude, OpenAI), agentic workflow design (multi-agent orchestration, tool-calling APIs -- framework-agnostic; fast ramp on LangChain/LlamaIndex)

ML & Data: Python (primary, production systems), PyTorch, SQL, data pipelines (Pandas, JSONL), vector databases (ChromaDB), RAG pipeline design, time series analysis (ARIMA), optimization (LP, KKT)

Infrastructure: AWS EC2 (deployment & serving), FastAPI (REST APIs), Docker (basic), Git / GitHub, MLOps (training pipelines, experiment tracking, lifecycle management)

Domain: Financial domain AI (LoRA fine-tuning, financial reasoning, equity research), regulated industry evaluation frameworks, high-stakes output quality design